

BRAC University
CSE340: Computer Architecture
Quiz#1

Duration : 20 Minutes

Total Marks : 15 Set: Abacus

ID:

Name:

Sec:

1. A program runs on an Intel Core i7 processor and consists of two segments:

Segment	No. of Instructions	Instruction Mix	CPI of Each Type
A	400,000	Arithmetic: 50% Memory: 30% Control: 20%	Arithmetic: 2, Memory: 4, Control: 3
B	600,000	Logical: 40% I/O: 40% Floating point: 20%	Logical: 3 I/O: 2 Floating point: 2

The CPU has a clock frequency of 1.536 GHz and a MIPS rating of 600.

- Find the average CPI for each segment. [3]
- Compute the overall average CPI for the entire program. [2]
- Find the total number of clock cycles required to execute the program. [2]
- Determine the execution time of the program in ms. [2]
- If the processor's frequency is increased by 20%, but the CPI of memory instructions also increases by 50% due to higher latency, find the new execution time in ms. [3]
- After a performance optimization upgrade, the processor's clock cycle time decreases, resulting in a MIPS rating of 700 for this program. Calculate the clock frequency (in GHz) required to achieve this MIPS rating. [2]
- Another new enhanced version of the processor claims a performance of 900 MIPS. However, the CPI for arithmetic instructions increases by 20% while the CPIs of all other instruction types remain the same. Determine whether the execution time for this program improves or worsens compared to the original 600 MIPS model. Show numerical comparison and calculate the percentage change in execution time. [4]
- After adding additional floating-point cores, the overall execution time of the program decreases by 10% compared to the original 600 MIPS model. Assuming that only the floating-point portion of the program benefits from this improvement, determine the speedup achieved for the floating-point operations. Show all necessary steps. [2]

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1. A program runs on an Intel Core i7 processor and consists of two segments:

Segment	No. of Instructions	Instruction Mix	CPI of Each Type
A	500,000	Arithmetic: 40% Memory: 35% Control: 25%	Arithmetic: 2, Memory: 3, Control: 3
B	500,000	Logical: 30% I/O: 50% Floating point: 20%	Logical: 4 I/O: 4 Floating point: 3

The CPU has a clock frequency of 2.0 GHz and a MIPS rating of 625.

- Find the average CPI for each segment. [3]
- Compute the overall average CPI for the entire program. [2]
- Find the total number of clock cycles required to execute the program. [2]
- Determine the execution time of the program. [2]
- If the processor's frequency is increased by 20%, but the CPI of memory instructions also increases by 50% due to higher latency, find the new execution time. Compare it with the original. [3]
- After a performance optimization upgrade, the processor's clock cycle time decreases, resulting in a MIPS rating of 750 for this program. Calculate the clock frequency (in GHz) required to achieve this MIPS rating. [2]
- Another new enhanced version of the processor claims a performance of 950 MIPS. However, the CPI for arithmetic instructions increases by 20% while the CPIs of all other instruction types remain the same. Determine whether the execution time for this program improves or worsens compared to the original 625 MIPS model. Show numerical comparison and calculate the percentage change in execution time. [4]
- After adding additional floating-point cores, the overall execution time of the program decreases by 5% compared to the original 625 MIPS model. Assuming that only the floating-point portion of the program benefits from this improvement, determine the speedup achieved for the floating-point operations. Show all necessary steps. [2]